Unfortunately, you can’t test the recovery process. The CloudWatch alarm monitors the health of the host system, which can be controlled only by AWS.

13.1.1 How does a CloudWatch alarm recover an EC2 instance?

Now that you’ve launched an EC2 instance with self-healing capabilities, it’s time to take a look at the details. The EC2 service checks the status of every virtual machine automatically. System status checks are performed every minute, and the results are available as CloudWatch metrics.

A *system status check* detects a loss of network connectivity or power, as well as software or hardware problems on the physical host. By default, AWS does not recover failed EC2 instances automatically. You can, however, configure a CloudWatch alarm based on the system status check to restart a failed virtual machine on another physical host.

Figure 13.2 shows the process if an outage affects a virtual machine, described here:

1. The physical hardware fails and causes the EC2 instance to fail as well.
2. The EC2 service detects the outage and reports the failure to a CloudWatch metric.
3. A CloudWatch alarm triggers the recovery of the virtual machine.
4. The EC2 instance is launched on another physical host.
5. The EBS volume and Elastic IP stay the same and are linked to the new EC2 instance.